

Fall 2026

Technical and Academic English
for Computer Science

Academic Research Process

Lecture #3

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Goals for Today

- > Understand the research process in computer science
- > Learn why and how research is published
- > Distinguish essential types of publications
- > Gain insights into peer-reviewing, the main quality control mechanism

Academic Research

What is research?

- > Typical researchers are easy to recognize



- > And in Computer Science?



Why is building pyramids still mysterious?



What is research?

- > An OECD publication¹ gives the following definition

“Research and experimental development (R&D) comprise **creative** and **systematic** work undertaken in order to **increase the stock of knowledge** - including knowledge of humankind, culture and society - and to **devise new applications** of available knowledge”

- > Further derives the following key characteristics

“The activity must be **novel** (aimed at new finding, not known yet), **creative** (based on original, non-obvious concepts/hypotheses), **uncertain** (outcome and/or successfulness unknown), **systematic** (planned & consciously managed; rigorous), **transferable** and/or **reproducible** (results could be reproduced)

¹ Frascati Manual 2015: Guidelines for Collecting and Reporting Data on Research and Experimental Development; doi:10.1787/9789264239012-en

What is research?

- > **The OECD definition is intentionally broad, and distinguishes several types of research based on their general goals and motivation**
 - Basic research (curiosity-driven research)
 - Applied research (application-driven research)
 - Experimental research (product-driven research)

- > **Note**
 - OECD's definitions have some (unavoidable) imprecision
 - There are other ways to define and classify research

How can we deliver the results?

- > **Writing: the most cost-efficient way**
- > **Environment, equipment, materials, ...**
 - Expensive to keep
 - Unavailable to deliver for some cases

Academic skills for the non-scientist

“I don’t want a career in research - why should I care?”

⇒ key aspects are important to everybody, in high-skilled jobs but also in life

1. Understanding science
2. Conducting research
3. Presenting results

Academic skills for the non-scientist

> Understanding science

- Be critical - tell facts from lies
- Understand how academic research works and what its weaknesses and limits are

> Conducting research

- Investigate a topic in detail
- Turn guesses & hopes into knowledge

> Presenting results

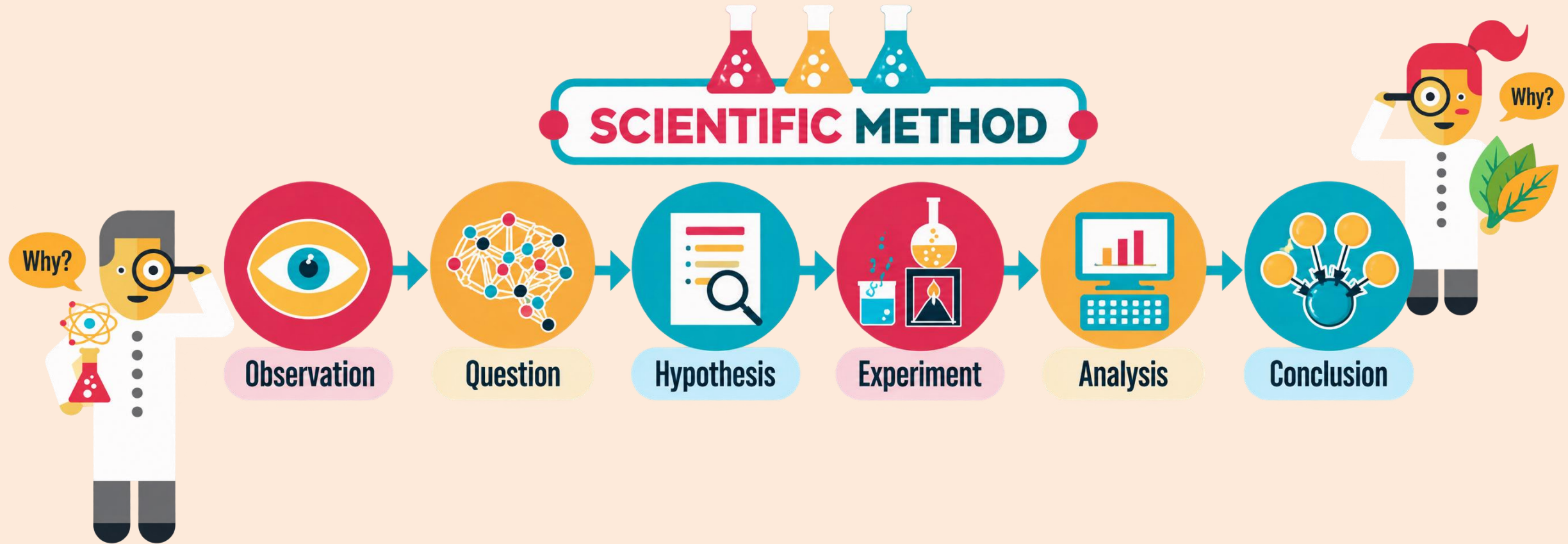
- Author reports, technical documents, etc.
- Present to audiences
- Your near future: seminar talks, project thesis, ...

Research process

How does systematic knowledge generation look like in computer science?

- > Individual research questions are often resolved by a sequence of steps
- > Zooming in
 - Each step may follow some distinct methodology (e.g., empirical evaluation, mathematical proof, etc.)
- > Zooming out
 - Research programs and whole research fields undergo long-term developments, characterized by asking many research questions and adjusting methodologies overtime

Research process



Research types

- > Theoretical research
- > Experimental research
- > Design-oriented research

Theoretical research

- > This is the predominant research process in mathematics and theoretical computer science

Theoretical research

1. Define objects of study
2. State a conjecture
3. Construct a proof
4. Discuss and interpret the results

Example: Is P the same as NP?

1. Define complexity classes and other relevant mathematical notions
2. Conjecture “ $P = NP$ ”
3. Proof by applying known mathematical identities
4. Discuss assumptions made in the proof (e.g., that $N = 1$ or $P = 0$); discuss further consequences

Experimental research

- > **This is the predominant research process in natural and social sciences**

Experimental research

1. Construct an initial theory (hypothesis, model)
2. Make a prediction based on the initial theory (model)
3. Design and carry out experiments to test the prediction
4. Analyze and compare outcome of the experiments with prediction

Example: Linux vs. Windows

1. Hypothesis: Linux is more user-friendly than Microsoft Windows
2. Predict that Linux users perform better on routine, everyday tasks
3. Ask groups of Linux and Windows users to configure and build kernel 5.0-rc7 on their system of choice; use think-aloud protocols and measure times
4. Discuss findings and possible biases in experimental setup

Design-oriented research

- > **This is the predominant research process in engineering**

Design-oriented research

1. State or derive requirements
2. Design a system
3. Implement and evaluate the system
4. Analysis results and draw conclusions about the system design

Example: Machine Learning (ML) on mobile devices

1. To train ML models on smartphones, we need energy-efficient algorithms
2. Design for a novel neuro-morphic, agile, embedded, adaptive, IoT-enabled single-layer perceptron
3. Implementation and evaluation w.r.t prediction quality and energy use
4. Discuss findings and consequences for the design of future ML architectures

Theory vs. Experimental vs. Design

- > **Real computer science research works will often combine aspects of several research processes**

Example: This could be the content of a single research work

1. Design a new algorithm and build a system using it
2. Hypothesize that this new approach is inherently faster than the old way of doing things
3. Give mathematical proof of better worst-case complexity properties
4. Design and carry out experiments to validate if those theoretical gains are relevant in practical settings
5. Discuss findings and results

- > **The design (of software, hardware, algorithms, mathematical theories, and other artifacts) is a part of most CS research, and can be combined with many methods**

The Academic Publication Process

Types of publications

- > Today, anybody can publish anything
- > So, what “counts” as a research publication?

Types of publications

The following basic types of text publications should be distinguished

- > **Formal research venues** : articles in journals and proceedings of research conferences, with established academic standards and rigorous quality control
- > **Informal research venues** : proceedings of workshops, meeting notes, etc.
- > **Monographs and collections** : books, including textbooks, and edited collections of invited research articles
- > **Theses** : texts written for obtaining an academic degree
- > **Technical reports** : self-published research papers that may not have undergone any quality control yet, but are usually archived and stable
- > **Other online texts** : blog posts and other web pages
- > **Fake publications** : Fraudulent or pseudo-scientific texts that try to look like research

Why (not) publish?

What makes people publish research results?

- > **Fame and recognition (publication pressure)**
 - Providing credits to the resulting ideas
- > **Seek exchange with other researchers and obtain feedback**
- > **Sharing knowledge with other researchers and society as a whole**
- > **Influencing opinion (promotion of own ideas & product)**

Why (not) publish?

What makes people not publish research results?

- > Lack of interest (no sufficient personal gain)
- > Retaining knowledge advantage
(industry and military research may be classified)
- > Negative effects for own agenda or business

Why (not) publish?

Conclusion

- > Some published research might be misleading, biased, irrelevant, and wrong and some highly significant and relevant research might not be published.

Ensuring quality and stability

Two key questions to count research publications

- > How has the quality of a work been ensured?
- > Is the work archived in some permanent way?

Ensuring quality and stability

	Quality control	Archiving
Formal research venues	Peer review, copy-editing	Publisher (typically library-indexed)
Informal research venues	Peer review/none	Publisher
Monographs and collections	None/friendly reviews, copy-editing	Publisher (typically library-indexed)
Theses	University process (varies widely)	University, libraries
Technical reports	None	Publisher or scientific archive
Other online texts	None	Publication site, archive.org
Fake publications	None	Publisher

Conferences vs. Journals in Computer Science

> **Conference proceedings**

- Fixed timeline (paper has to be accepted before event happens)
- Length restrictions (to allow timely review)
- Short time to publication (typical: two months submission - acceptance)

> **Journal articles**

- Open timeline (reviewers can ask for minor or major changes)
- Usually no length restriction (to allow detailed, rigorous presentation)
- Possibly significant time to publication

Conferences vs. Journals in Computer Science

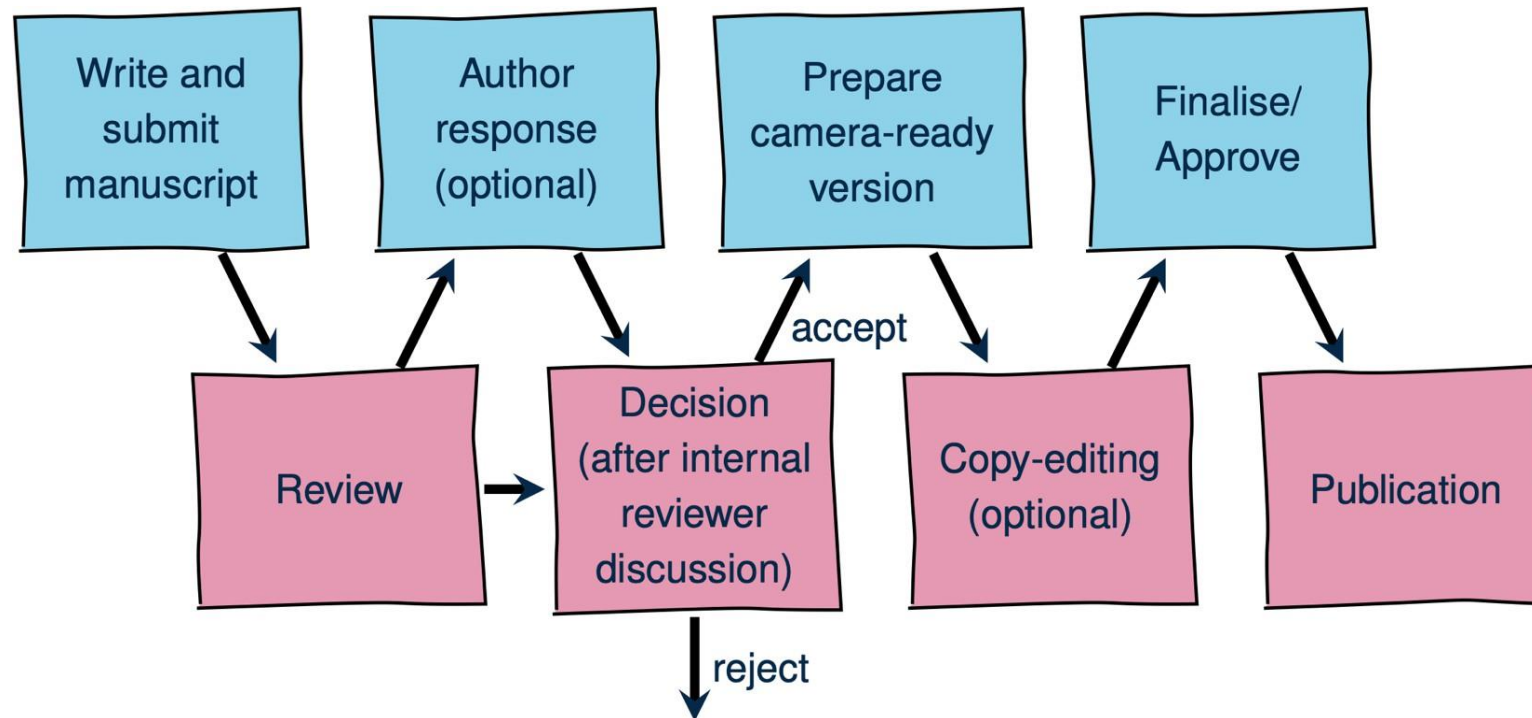
> Workshop, Posters, Short papers

- “Workshop” in computer science are often mini-conferences for preliminary works
- Some conferences also offer second-tier publication formats that do not get full articles in the proceedings (poster presentations, short papers, etc.)
- Emphasis of such events is on exchange; little relevance as publication venues

⇒ Note: other academic fields have completely different publication cultures!

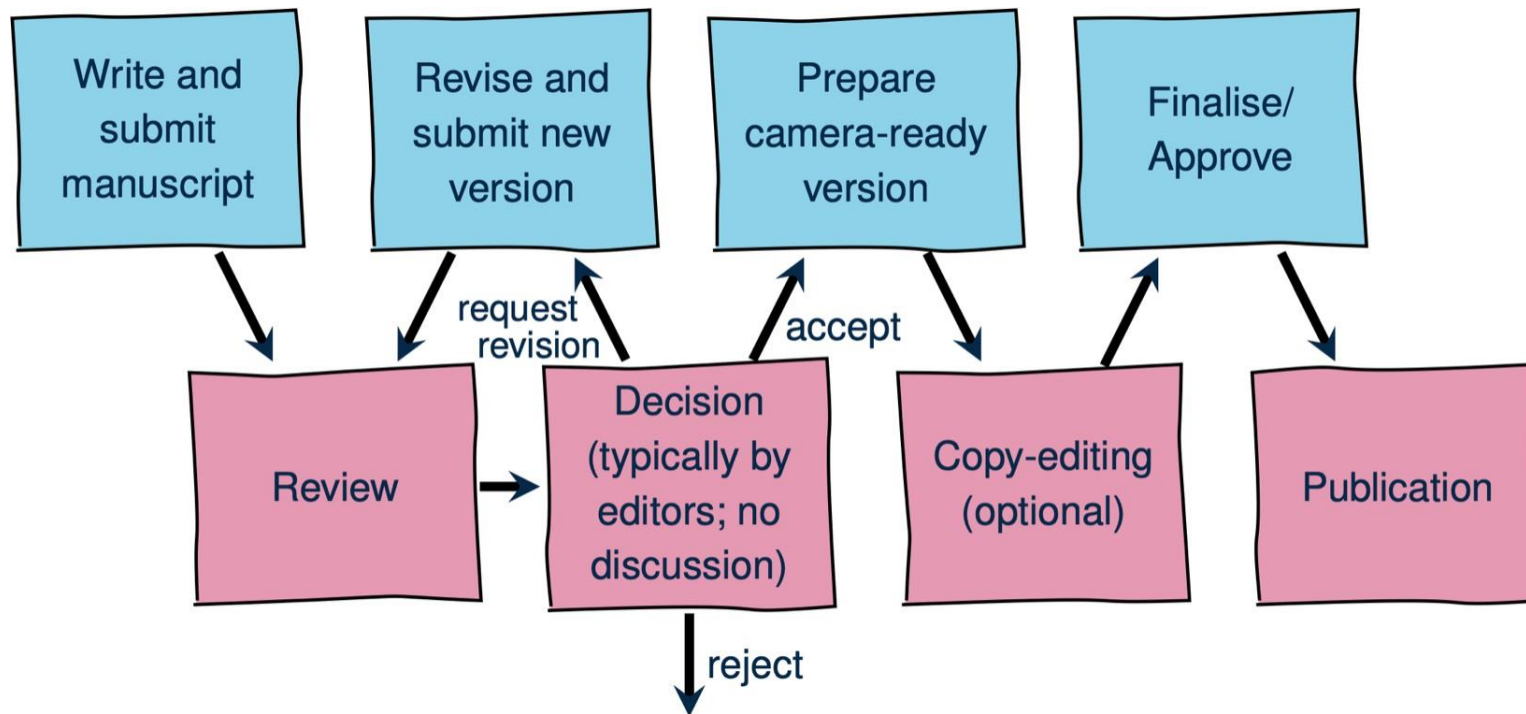
Publication process in computer science

- > Most computer science research is published at (small and large) conferences/workshops
- > The general publication process for conferences is as follows:



Publication process in computer science

- > Journals adopt a slightly different process, since they publish more frequently and are therefore not bound to a firm timeline for acceptance
- > The general publication process for journals is as follows:



Peer review vs. copy-editing

- > **The predominant quality control mechanism in research is peer review**

Peer review

- Manuscripts are submitted to a publication venue
- An editor/program chair asks experts to review the submission
- Based on the experts' opinions, the editor/program chair decides if the submission can be accepted

Peer review vs. copy-editing

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- > **This scientific quality control is different from the copy-editing done by some publisher**

Copy-editing

- Accepted manuscripts are finalized and sent to the publisher
- A trained copy-editor checks language and formatting issues
- Comments are sent to the authors or implemented directly by the copy-editor (but the authors should always give final approval)

Peer review

- > **The most widely used method of quality control across all of academia**
- > **Several basic models**
 - Single-blind
 - Double-blind
 - Non-blind (open)

Peer review: Single-blind

- > Reviewers are not known to the authors
- > Most common model today
- > **Motive:** protect reviewers from authors (who might be unhappy with verdict), prevent authors from trying to influence reviewers privately

Peer review: Double-blind

- > Even reviewers do not know authors' identity
- > Increasingly popular, but not always practical
- > esp. journal articles must discuss own prior works, thus revealing author identity in the context
- > **Motive:** increase changes for outsider authors, who might be reviewed more skeptically than the “big names” when presenting new ideas

Peer review: Non-blind

- > Reviewers and authors know each other's identity
- > Implemented by some journals and conferences
- > **Motive:** make reviewers more accountable for their reviews; increase reviewing process transparency

How peer review works

> How are reviews organized?

- Journals: the managing editor invites experts to provide a review
- Conferences: the programmed chair recruits a program committee (PC) upfront; each PC member will be assigned several submissions to review
- Large conferences: sometimes use hierarchies of reviewers (extra roles include area chairs, track chairs, and senior PC members)
- Book projects: the editors organize reviews in whatever way they see fit

⇒ Note: Reviewing is a community service and usually not paid for!

How peer review works

> What are the results of a review?

- **Journals:** usually “accept”, “accept with minor revision”, “request major revisions”, “reject with suggestion to resubmit”, “reject”
- **Conferences :** “Accept” or “reject”, sometimes also “accept with short paper” or similar. Reviews use more fine-grained scoring systems such as “Strong accept”, “accept”, “borderline”, “weak reject”, “reject”, “strong reject”

> How many reviews per submission?

- Usually three or more; rarely just two or even just one; sometimes none (“desk reject”)

How peer review works

> Who can be a reviewer?

- Any qualified expert
- Should have own research and publication experience (someone who never wrote a journal paper is not in the best position to tell others how to do it)
- Must not have **a conflict of interest**

- Author is a former Ph.D supervisor, former Ph.D student, family member, close friend
- Recently (past three years): collaborations with author, working in the same organizational unit
- Conflicting commercial or academic interest
- Any other circumstance that prevents somebody from giving a fair and unbiased review

Reviewing criteria

- > Especially conferences often ask for reviews to evaluate several criteria for better comparability of submissions

Reviewing criteria

Typical criteria includes

- > **Relevance:** Does the contribution fit the conference/journal? Is it relevant to the research area it was submitted in?
- > **Significance:** Are the results significant (large enough)? Does it advance our knowledge a lot?
- > **Originality:** Is the work novel (new results, new methods, etc.)? Also compared to prior publications by the same authors
- > **Verifiability:** Is the work reproducible?

Reviewing criteria

Typical criteria includes

- > **Correctness:** Are the claims likely to be true? Are the proofs free of errors? Are the experimental designs sound? Are the conclusion valid?
- > **Presentation:** Is the paper readable and clear?
- > **Related work:** Does the work clarify how it compared to previous works in this area? Are all relevant references cited?

⇒ Reviewers are often asked to rate and comment on each dimension

Summary

- > **Several different research processes are common in computer science**
- > **Publications play a key role in making research results known**
- > **A rigorous and trustworthy quality control mechanism is essential to ensure that research publications are useful**
- > **Peer review is the most widely used quality mechanism**
- > **Publication is not a definite certificate of high quality – critical thinking is needed**

In Lecture #3

- > **Gathering information in research**
- > **Finding the most relevant literature and experts**
- > **Reading academic papers**